

Project Introduction

Instructor

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State of LLMs

Rise of Open Source Models:

Open-source models such as Meta's Llama, Alibaba's Qwen, and MistralAI's models are performing comparable to proprietary models like OpenAI's ChatGPT and Anthropic's Claude in multiple LLM benchmarks.



State of LLMs

Inference Resource Requirements:

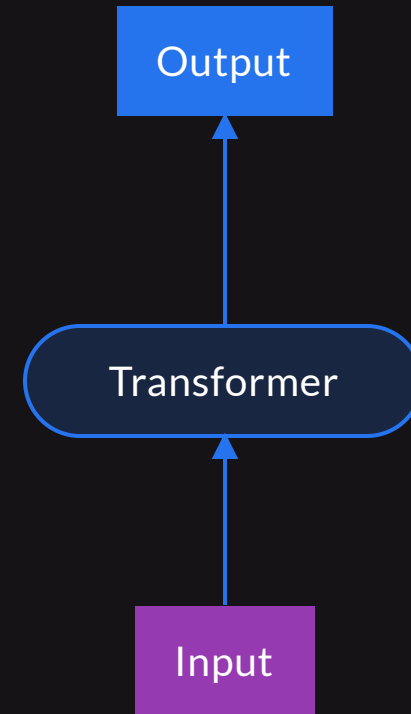
- For inference, 1 billion parameters at 32-bit floating point (FP) precision require 4 GB of VRAM
- Calculation 32-bit FP
 - 32-bit FP = 4 Bytes
 - 4 Bytes $\times 10^9$ (1B Params) = 4 GB VRAM
- Llama3 70B inference cost: $70 \times 4 = 280$ GB of VRAM
- **AWS p4d.24xlarge**
 - 96 vCPUs
 - 1237 GB
 - 8 x A100 GPUs = 320 GB
 - On-demand = 32.77 USD/hr
 - Per month cost = \$23594.4 = 19.7 lacs



Full Fine-Tuning

Training GPU usage:

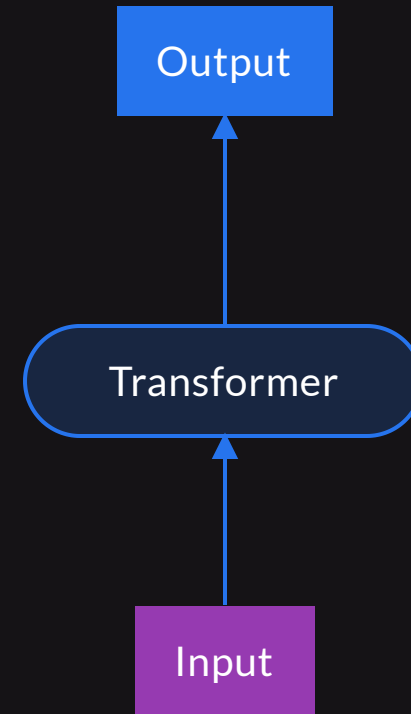
- For 1 parameter in transformer weights
 - 2 x Optimizer states
 - 1 x Gradients
 - 2 x Activations and temp memory
- 4 GB weights = $4 \times 6 = 24$ GB @ 32FP



Full Fine-Tuning

Issues with full fine-tuning:

- Large memory size
- Maintaining and deploying multiple models
- Catastrophic forgetting on fine-tuning leading to poor results on general tasks



Parameter Efficient Fine-Tuning

Fine-tuning

- Fine-tuning techniques like SFT, RLHF, PPO
- We will fix to SFT Supervised Fine Tuning

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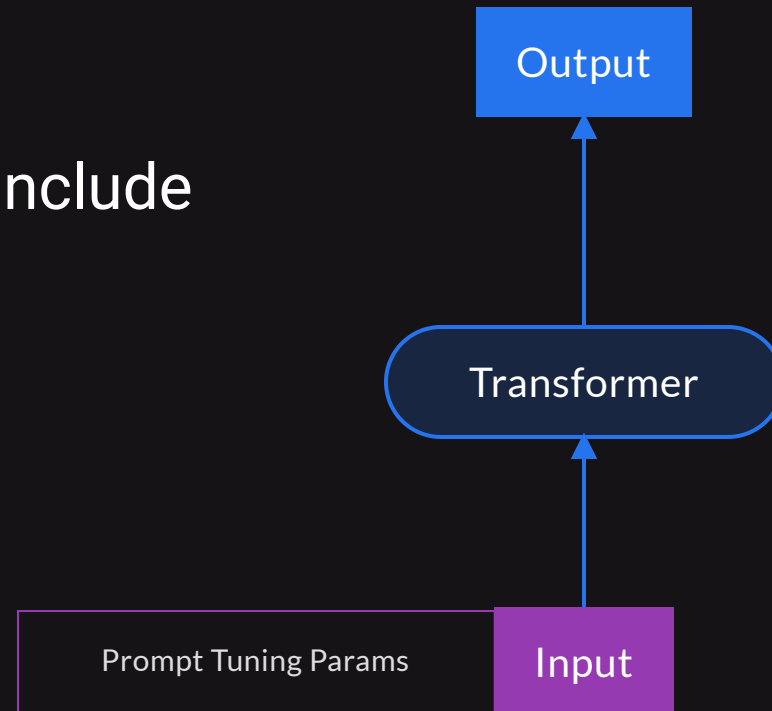
In SFT

- All or some weights kept frozen
- Either small number of layers are trained
- Or new layers are trained and added later

Parameter Efficient Fine-Tuning

Popular techniques include

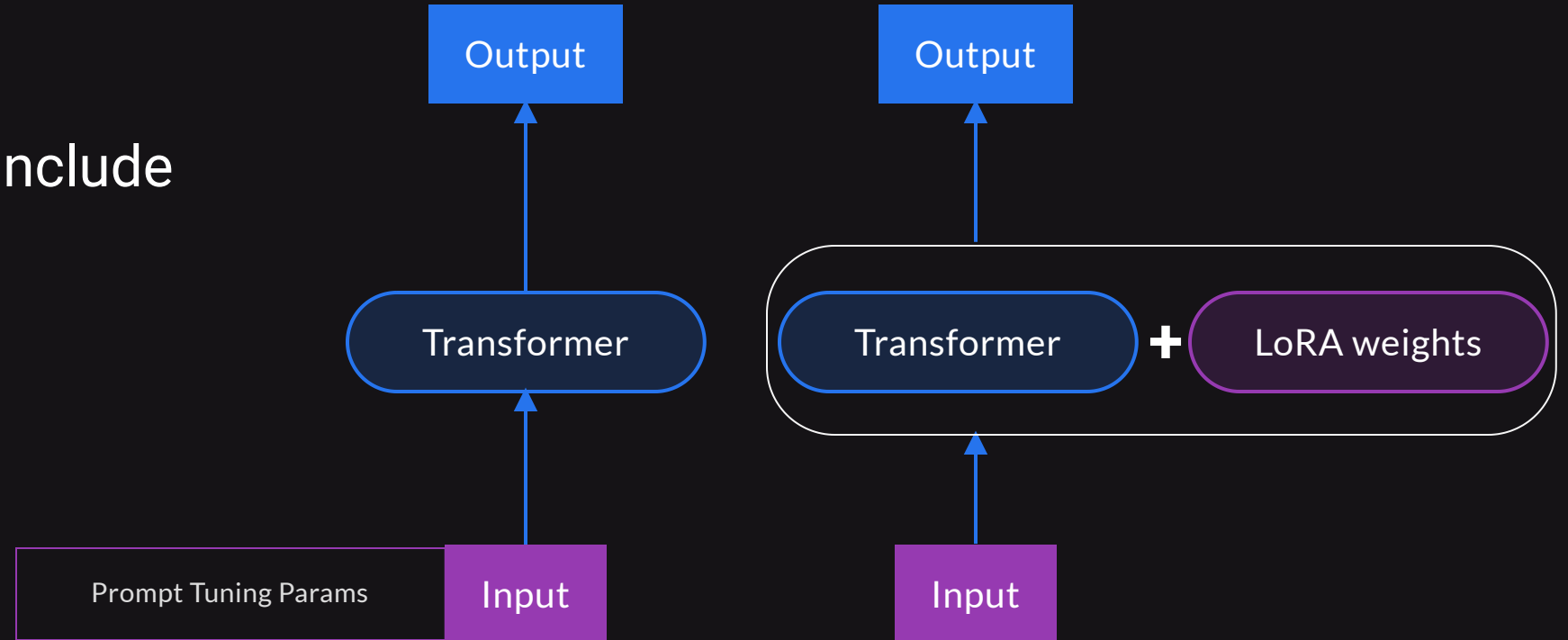
- Prompt Tuning



Parameter Efficient Fine-Tuning

Popular techniques include

- Prompt Tuning
- LoRA



LoRA: Low-Rank Adaptation

LoRA training

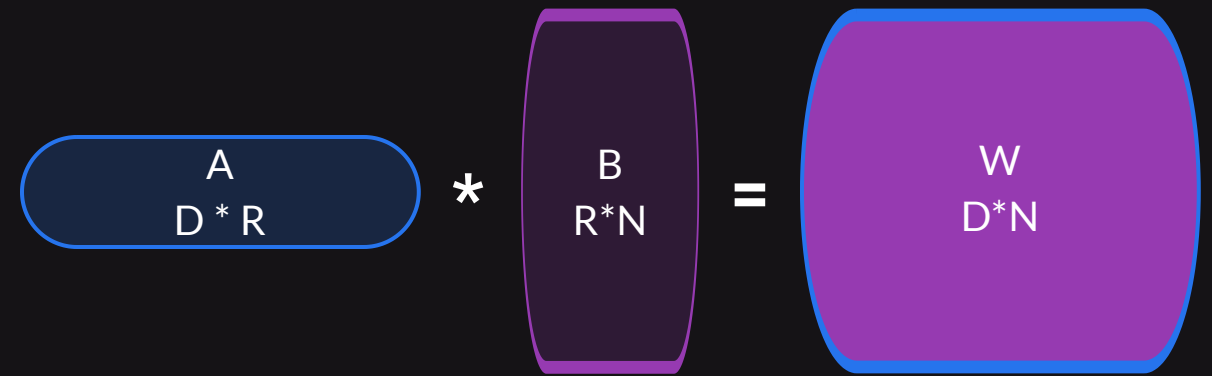
- Reparameterize by adding low rank decomposition matrices to different layers of the transformer
- Subset of params to fine tune - mostly the later layers as they contain most of the combined knowledge
- By research at Microsoft a value of rank between 4-10 gives good performance

$$\begin{matrix} A \\ D * R \end{matrix} * \begin{matrix} B \\ R * N \end{matrix} = \begin{matrix} W \\ D * N \end{matrix}$$

LoRA: Low-Rank Adaptation

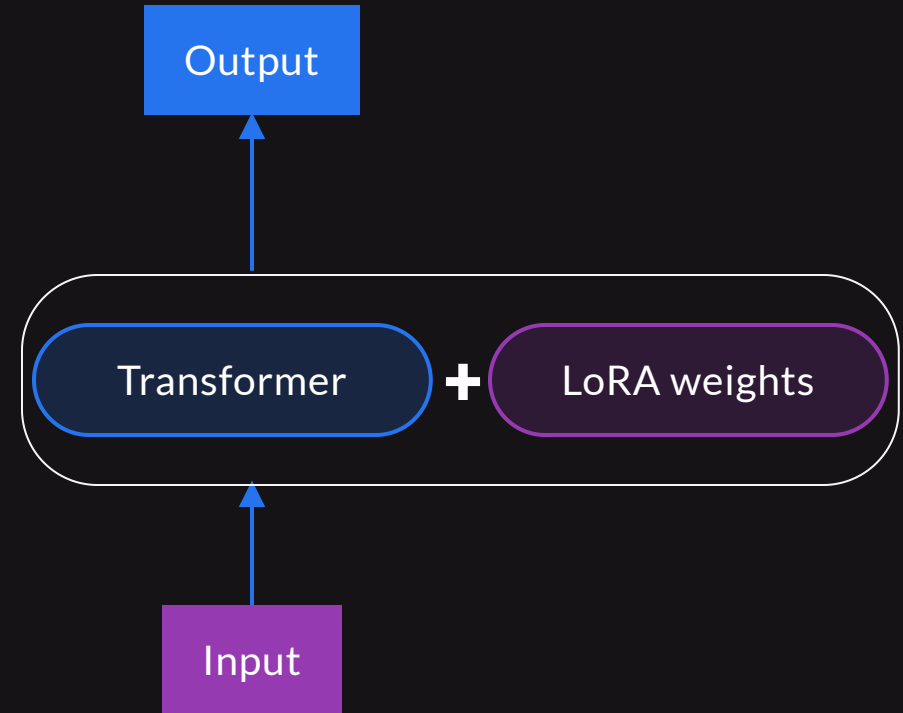
Advantages of LoRA

- Single GPU training for 7-10 billion parameter models
- Scores are comparable to full-fine tuned models
- Many LoRA adapters can be trained for several downstream task with the same base model
- No added latency due to fused weights at runtime

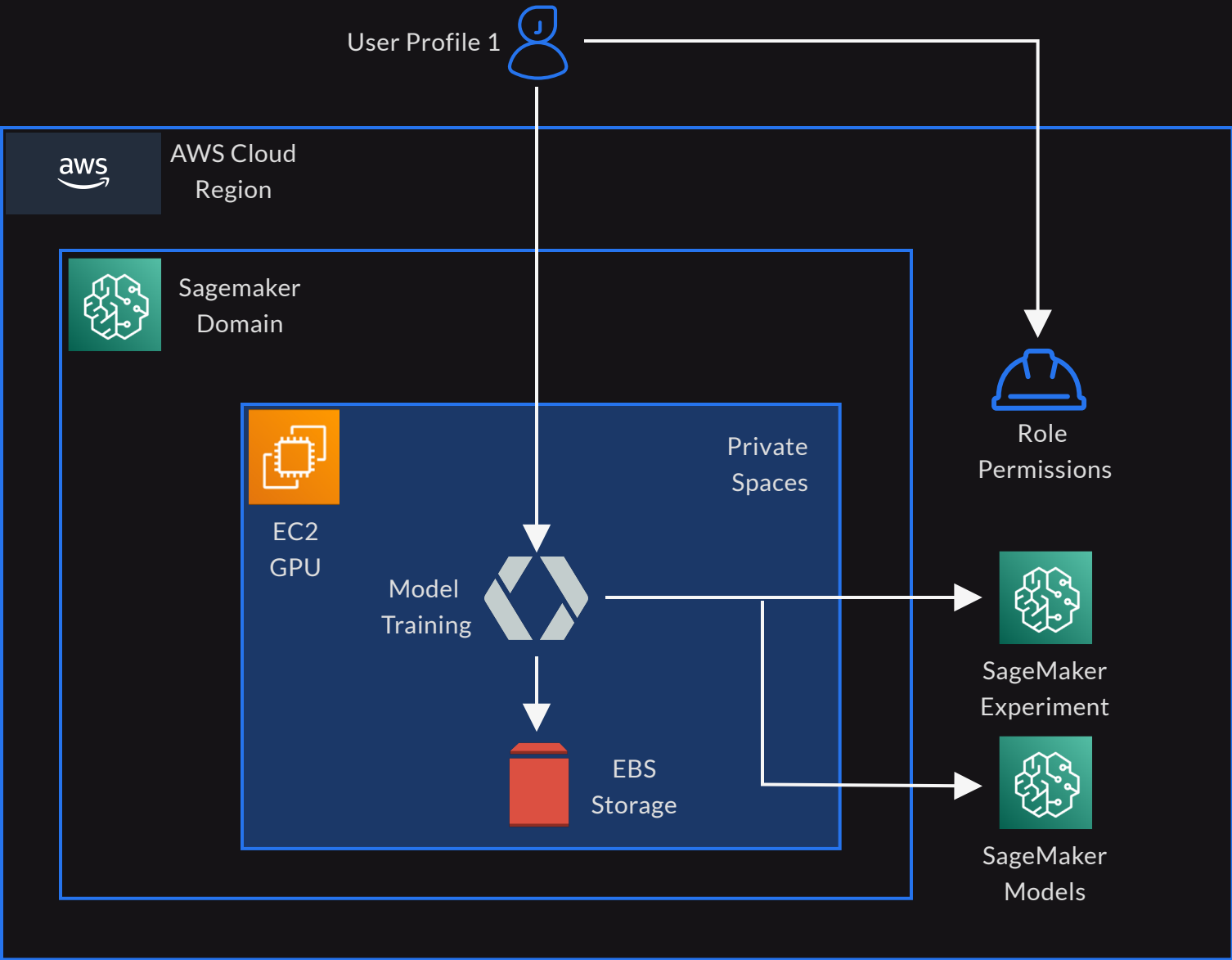


Project Overview

- Prepare dataset for fine-tuning
- Llama3 as base model
- Fine tuning using Hugging Face libraries
 - Quantization to reduce model size
 - QLoRA based supervised fine-tuning (SFT)
- Evaluate the LLMs output



SageMaker Infrastructure



Thank You
